

Sarthak Bichhawa

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Summary

Senior Data Scientist (9+ years) driving product analytics, metrics frameworks, and AI-assisted decision-making—governed reporting, revenue attribution, and executive customer-health programs—with >95% reporting toil removed and ~\$45M/month in recovered revenue.

Experience

Microsoft, Redmond, WA | *Sr. Data & Applied Scientist*

Nov 2020 – Current

Metrics & Customer-Health Programs:

- Drove the **Mission-Critical Customers (MCC) health program** and a unified customer-health view across **8+ engineering/support teams**, cutting executive/customer readiness insight from **3–5 days to near real-time**—with ~50% fewer >10-day aging issues and **~16% faster time-to-mitigate**.
- Helped reverse churn/downtrend risk for **10+ critical customers**—each representing multi-millions in monthly Azure Storage revenue—by accelerating early detection and action on customer-impacting issues.

Revenue Attribution & Business Reporting:

- Recovered and sustained **~\$45M/month in influenced partner revenue** (at <1% variance vs Finance, ~4× partner coverage) by rebuilding a 20+ component partner-attribution pipeline with **multi-partner credit** replacing last-writer-wins.
- Collapsed monthly business-review reporting from a multi-day manual process to a governed **~5-minute pipeline** (>95% effort reduction), now in regular leadership use; shipped Files/Disk/Blob business-insights dashboards on a curated dataset layer.

Data Quality & Governed Metrics:

- Built and operated **large-scale telemetry pipelines and governed metrics**, applying statistical analysis and anomaly detection to validate data quality and turn ambiguous signals into decision-ready insights.
- Caught silent upstream data regressions—including an **~88% row-collapse**—before they reached Finance by building a self-service **data-quality monitoring service** with anomaly detection and hourly table-health profiling over 47 datasets.

Product Impact at Scale:

- Shipped **15+ Azure Advisor recommendations** adopted across hundreds of thousands of customer accounts to improve cost and performance, partnering with engineering and support to validate and operationalize.

Self-Serve AI Insights:

- Architected an enterprise **LLM agent** that turns natural-language questions into governed live Kusto/ADX queries and PM-ready narratives, cutting customer-health time-to-insight from **days to <2 minutes** across 13 metric domains.

Walmart Labs, Bentonville, AR | *Senior Data Scientist*

Feb 2019 – Nov 2020

- Delivered **~7% category revenue lift** across 50 stores with an end-to-end **planogram optimization** system, cutting planning cycles from weeks to hours.
- Led **4 DS/analysts + 2 interns** to productionize a repeatable item-level placement workflow; earned a Spot Award.

Integral Ad Science, New York, NY | *Data Scientist Intern*

June 2018 – Aug 2018

- Built a PySpark contextual text-classification pipeline processing **~1M URLs/day across 846 categories** (88% accuracy); improved label quality via correlation analysis.

Mu Sigma Business Solutions, Bengaluru, India | *Data Scientist*

Oct 2015 – July 2017

- Estimated **customer lifetime value (CLTV)** with probabilistic/regression models; improved competitive reporting by unifying product IDs via fuzzy matching.

Education

University of Illinois at Chicago (UIC), Chicago, IL | *MS-Business Analytics*

Aug 2017 – Dec 2018

SGSITS, Indore, India | *B.E. in Computer Engineering*

Aug 2011 – July 2015

Skills

- **Analytical Techniques:** Experiment & metric design, causal inference, statistical modeling/validation, clustering & segmentation, anomaly detection, generative AI / RAG analytics
- **Analytical Languages & Tools:** Python, SQL, Azure Data Explorer (Kusto), PySpark, R, FastAPI, Power BI, Tableau, Azure DevOps, Git
- **Functional Areas:** Product analytics & metrics frameworks, partner/revenue attribution & business reporting, customer insights & health programs, XFN data strategy, self-serve LLM insights tooling, production ML